

The Optimization Landscape

When local information tells the global truth

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Mathematical Foundations for AI · IIT Gandhinagar

Two optimization problems

Linear and nonlinear model fitting

Both of these minimize a loss — L14 taught us that's what “fitting” is:

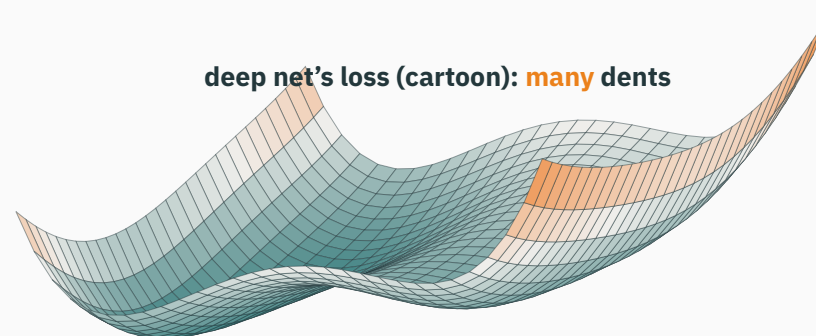
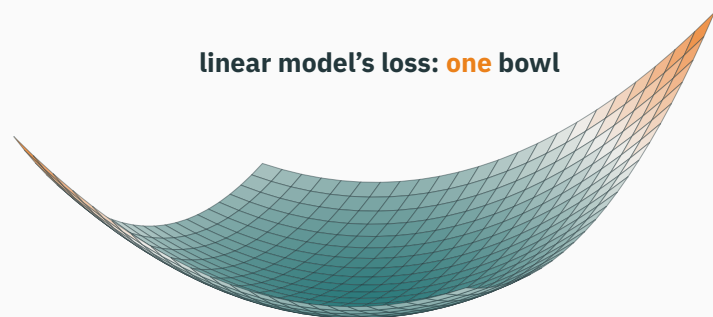
	fit a linear model	train a deep network
the code	<code>np.linalg.lstsq(X, y)</code>	<code>loop of <code>loss.backward()</code> ; <code>opt.step()</code></code>
knobs to tune	none	learning rate, schedule, warmup, init, clipping, batch size, ...
rerun on the same data	same answer, every time	depends on the seed
failure modes	—	divergence, plateaus, nan at step 617 (L2!)

Both tasks **minimize a function of the parameters**, but the two objectives have different geometry.

Linear and nonlinear model fitting

Question. Which differences come from the algorithm, and which come from the function being minimized?

The difference is the **shape of the surface** each one walks on:



- Left: the objective has one basin; every local minimum is global
- Right: the objective can have several valleys, ridges, and saddle points

For a convex objective, every local minimum is a global minimum.

From estimation to optimization

L14 expressed parameter fitting as minimization:

$$\hat{\theta} = \arg \min_{\theta} \underbrace{\text{NLL}(\theta)}_{\text{MSE, BCE, CE, ...}}$$

L14: “Fitting = minimizing NLL. Module 4 is the art of walking downhill”

Module 4 studies this minimization problem in six stages:



This lecture develops the geometry used by the remaining optimization lectures.

Learning outcomes

By the end of this lecture you will be able to:

1. Use the optimizer's vocabulary: **objective, minimizer, stationary point, local vs global**.
2. Classify flat spots by the Hessian's **eigenvalue signs** — L9's gallery, now on the job.
3. Test a **set** for convexity with segments, and a **function** with chords.
4. Certify convexity fast: $f'' \geq 0 / H \succeq 0$, plus the **sum, max, affine-composition** rules.
5. Prove ★ that every local minimum of a convex function is global.
6. Prove **least squares is convex** via $H = 2X^\top X \succeq 0$ (★) — and say what deep nets give up.

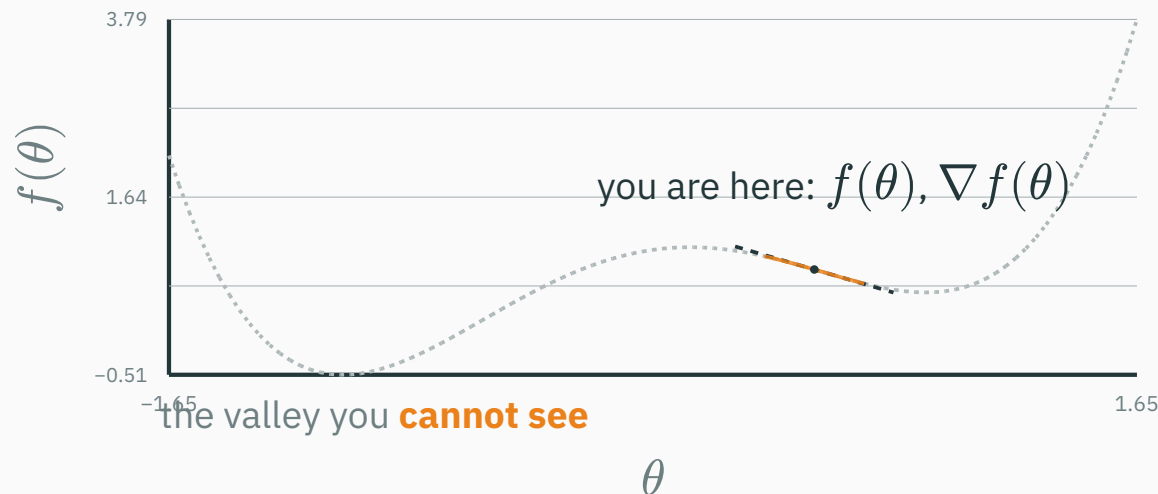
**Local information available to
an optimizer**

The problem, stated once for the whole module

$$\min_{\theta \in \mathbb{R}^d} f(\theta)$$

- f : the **objective** — for us, almost always a loss / NLL (L14)
- a **minimizer** θ^* : an input where the smallest value is reached — $\theta^* = \arg \min_{\theta} f(\theta)$; the **minimum** $f(\theta^*)$ is the value there
- maximizing is the same craft upside-down: $\max f = -\min(-f)$ — L14 already negated log-likelihoods for exactly this reason, so **everything from now on is a minimization**

The optimization problem is $\min_{\theta} f(\theta)$; the following lectures differ in how they use information about f .



At a current parameter value θ , an optimizer can evaluate three kinds of local information:

probe	what it is in ML	cost
$f(\theta)$	run the model, read the loss	cheap
$\nabla f(\theta)$	backprop (L8–L11)	cheap — one backward pass

probe	what it is in ML	cost
$H(\theta)$	curvature matrix (L9)	d^2 entries — rarely affordable

Why not just look everywhere?

Because θ lives in $\mathbb{R}^d$, and d is not 2. Grid-search with a modest 10 values per coordinate:

model	d	grid evaluations 10^d
line $wx + b$	2	$10^2 = 100$ — trivial
tiny MLP	10^3	10^{1000} — no.
GPT-class model	10^{11}	$10^{10^{11}}$ — the observable universe has $\sim 10^{80}$ atoms

Exhaustive search is impossible at ML scale; practical methods use local evaluations of f , ∇f , and sometimes H .

Local derivatives describe a neighbourhood. Convexity is the additional global property that makes those local statements sufficient.

Taylor models use local information

L9 built “the formula every optimizer lives on” — today it starts living:

$$f(\theta + \delta) \approx \underbrace{f(\theta)}_{\text{probe 1}} + \underbrace{\nabla f(\theta)^\top \delta}_{\text{probe 2: the local plane}} + \underbrace{\frac{1}{2} \delta^\top H(\theta) \delta}_{\text{probe 3: the local bowl or saddle}}$$

- L17 trusts the first two terms for one small step — **gradient descent**
- L18 trusts all three and jumps — **Newton’s method**
- Today’s question is prior to both: when do **local** conclusions (“it’s flat here”, “it curves up here”) say anything about the **global** landscape?

**Stationary points: where the
search can stop**

Stationary points

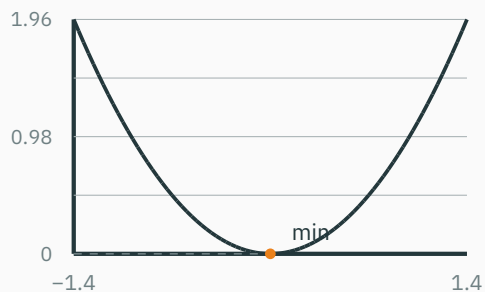
While $\nabla f(\theta) \neq \mathbf{0}$, the Taylor plane is tilted; a sufficiently small step against the gradient lowers f (L17 makes this precise).

it can only park where $\nabla f(\theta) = \mathbf{0}$ — a **stationary point**

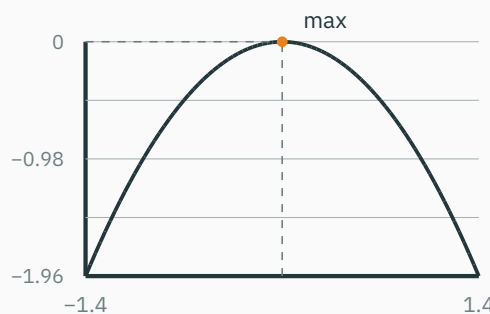
- Also called a **critical point**; the first-order model is flat there
- But **flat is not the same as bottom** — L7 met smiles, frowns and x^3 's fake-out in 1-D

The gradient test finds candidate optima. The Hessian distinguishes minima, maxima, and saddle points.

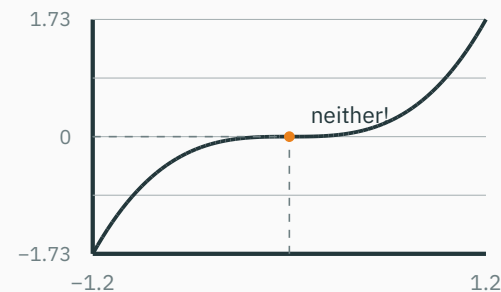
Classifying stationary points in one dimension



$f'' = 2 > 0$ – smile



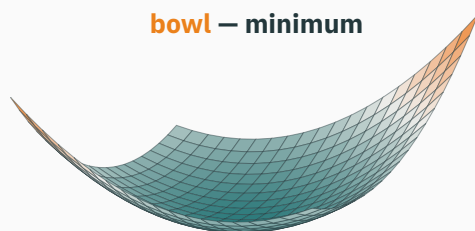
$f'' = -2 < 0$ – frown



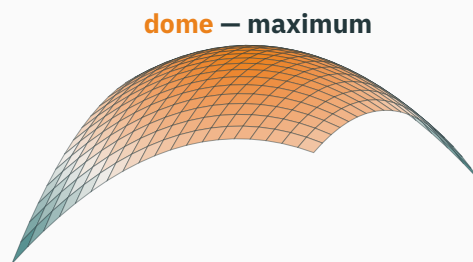
$f'' = 6x = 0$ – inconclusive

All three have $f'(0) = 0$. The **second** derivative tells them apart – except when it's 0, where the test goes silent (x^3 : neither a min nor a max).

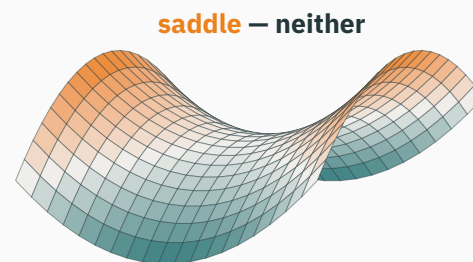
Hessian eigenvalues classify the local shape



$$\lambda = 3, 1 - \text{all} > 0$$



$$\lambda = -1, -3 - \text{all} < 0$$



$$\lambda = 2, -2 - \text{mixed signs}$$

Curvature became a matrix in L9; its **eigenvalue signs** are the shape. (Eigenvalues above: computed live by the in-slide solver.)

At a stationary point: H 's eigenvalue signs name the flat spot — all + min, all – max, mixed = saddle.

The Hessian-eigenvalue test

eigenvalues of H at $\nabla f = 0$	classification	descent implication
all $\lambda_i > 0$ (PD)	local minimum	park — locally, done
all $\lambda_i < 0$ (ND)	local maximum	any direction escapes
mixed signs (indefinite)	saddle	escape along a $\lambda < 0$ eigenvector
some $\lambda_i = 0$	test is silent	look beyond second order (x^3 , troughs)

Hand tests for 2×2 still apply (L9): $\det H < 0 \rightarrow$ saddle; $\det H > 0$ — read the trace.

This test is local. Positive Hessian eigenvalues certify a local minimum, not a global one.

Worked example: two stationary points

$$f(x, y) = x^3 - 3x + y^2 \quad \nabla f = \begin{pmatrix} 3x^2 - 3 \\ 2y \end{pmatrix} = \mathbf{0} \Rightarrow (x, y) = (\pm 1, 0)$$

Hessian: $H = \begin{pmatrix} 6x & 0 \\ 0 & 2 \end{pmatrix}$ — depends on where you stand (curvature is local):

- At $(1, 0)$: $H = \begin{pmatrix} 6 & 0 \\ 0 & 2 \end{pmatrix}$, $\lambda = 6, 2$ — both positive → **local minimum**
- At $(-1, 0)$: $H = \begin{pmatrix} -6 & 0 \\ 0 & 2 \end{pmatrix}$, $\lambda = -6, 2$ — mixed → **saddle**

And since $f(x, 0) = x^3 - 3x \rightarrow -\infty$ as $x \rightarrow -\infty$: that “local minimum” is **not** global — nothing is. Flat-spot analysis alone can never tell you this.

Classify both points in NumPy

```
import numpy as np

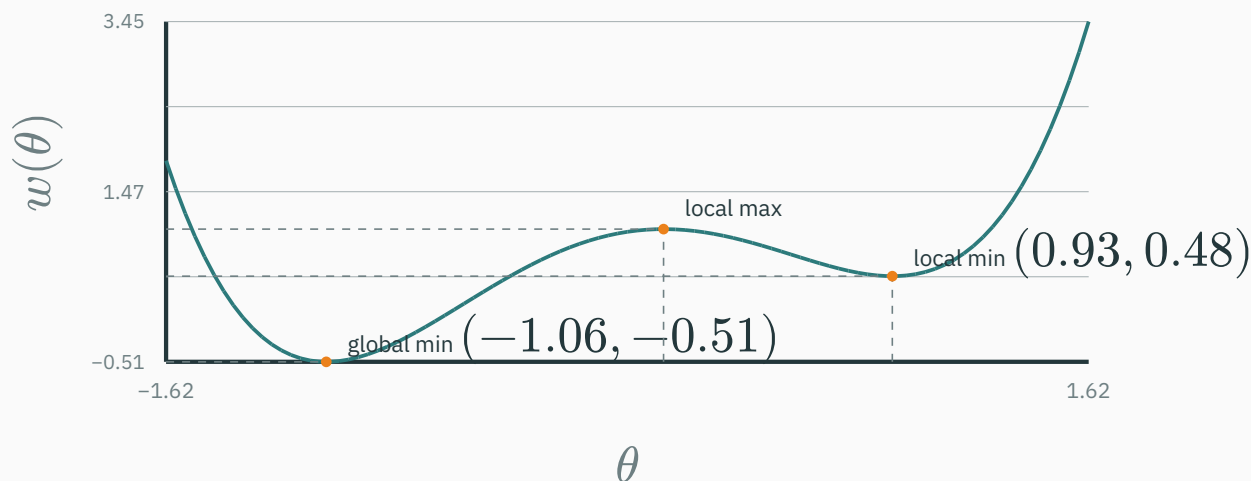
def hess(x, y):
    # for f = x**3 - 3*x + y**2
    return np.array([[6*x, 0.0], [0.0, 2.0]])

for p in [(1, 0), (-1, 0)]:
    lam = np.linalg.eigvalsh(hess(*p))
    # symmetric -> real, sorted
    kind = "min" if lam[0] > 0 else ("max" if lam[-1] < 0 else "saddle")
    print(p, lam, kind)
# (1, 0) [2. 6.] min
# (-1, 0) [-6. 2.] saddle
```

`eigvalsh` is the same symmetric-eigenvalue calculation used in L9; here the signs determine the local classification.

A local minimum need not be global

Consider the non-convex function $w(x) = (x^2 - 1)^2 + x/2$. Every marked value below is computed in-slide using Newton's method on $w' = 0$:



- **Local minimum**: no better point in some ball around it. **Global minimum**: no better point anywhere.
- A walker started at $\theta = 1.5$ parks at 0.93 — flat, curving up, **fully convinced** — and is wrong by 1.00 in loss.

A local minimum need not be global

The values of f , ∇f , and H at 0.93 do not reveal the lower minimum at -1.06 .

Checkpoint: what do local derivatives certify?

QUESTION

An optimizer stops at $\hat{\theta}$ on a smooth, **unknown** $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, with $\nabla f(\hat{\theta}) = 0$ and $H(\hat{\theta})$ having eigenvalues 2 and 5. What is **certain**?

A. $\hat{\theta}$ is a local minimum — and that is all we can certify

B. $\hat{\theta}$ is the global minimum

C. $\hat{\theta}$ is a saddle point

D. f is convex

Correct: A — a local minimum — and nothing more

Why: Both eigenvalues positive $\rightarrow$ PD $\rightarrow$ a genuine local min (L9). But **global** (B) is a claim about all of $\mathbb{R}^2$, and local probes at one point can't reach that far — our two-valley mascot just demonstrated it. (D) reverses today's logic: curvature at **one** point never certifies convexity — x^3 has $f''(1) = 6 > 0$ yet isn't convex. Turning “local min” into “global min” needs a **property of f** , not a better probe. Enter convexity.

**Convex sets: segments that
never leave**

The fix must be a property of shapes

What we want is a **certificate**:

$$\underbrace{\text{local conditions}}_{\nabla f=0, H \succeq 0} + \underbrace{\text{a property of } f \text{ as a whole}}_{?} \Rightarrow \text{global claim}$$

That property is **convexity**, and it comes in two pieces, buildable in one lecture:

- **convex sets** — regions where straight segments never leave (this section)
- **convex functions** — surfaces whose chords never dip below the graph (next)

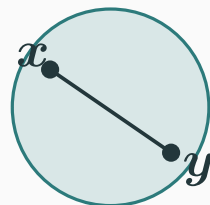
Sets first because functions need somewhere to live: the chord test only makes sense if the segment between two inputs stays inside the domain.

The segment test

A set $C \subseteq \mathbb{R}^d$ is **convex** if for all $x, y \in C$ and every $t \in [0, 1]$:

$$tx + (1 - t)y \in C$$

- $tx + (1 - t)y$ is a point **on the segment** from y to x — t is the slider
- Convex = “walk between any two members and you never step outside”



One definition, one picture: no shortcut between members ever leaves the club.

Numbers: walking the segment

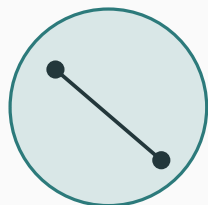
Take $x = (4, 1)$ and $y = (0, 3)$. The mix $tx + (1 - t)y$, by hand:

t	0	1/4	1/2	3/4	1
mix	(0, 3)	(1, 2.5)	(2, 2)	(3, 1.5)	(4, 1)

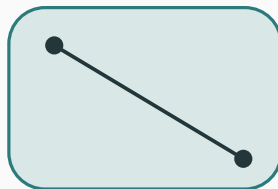
- $t = 0$ sits at y , $t = 1$ at x ; in between, a straight walk — each coordinate is mixed independently
- The same expression with weights $(t, 1 - t)$ will reappear within minutes as the **chord** of a function — same algebra, one dimension up

Vocabulary you'll meet in Boyd: a **convex combination** is a mix with nonnegative weights summing to 1 — segments are the 2-point case.

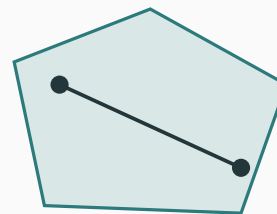
Examples of convex and non-convex sets



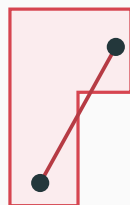
disk ✓



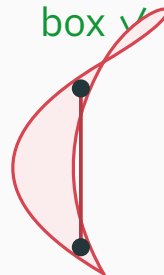
box ✓



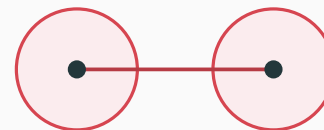
polytope ✓



a notch ✗



a crescent ✗



two islands ✗

One escaped segment is enough to convict: convexity is a **for-all** claim, non-convexity needs one witness.

The convex sets ML lives on

set	where you meet it	convex?
$\mathbb{R}^d$ — all of parameter space	unconstrained training	✓
box $[0, 1]^d$	valid pixel intensities	✓
half-space $\{x : a^\top x \leq b\}$	one side of a flat wall	✓
norm ball $\{\theta : \ \theta\ \leq r\}$ (L3)	weight constraints, regularization	✓
simplex $\{p : p_i \geq 0, \sum_i p_i = 1\}$	probability vectors — softmax outputs live here	✓ (L19 optimizes over it)
sphere $\{x : \ x\ = 1\}$ — the skin only	normalized embeddings	✗

The sphere fails beautifully: mix two antipodes at $t = 1/2$ and land at 0 — the center, off the skin. **Ball convex, shell not.**

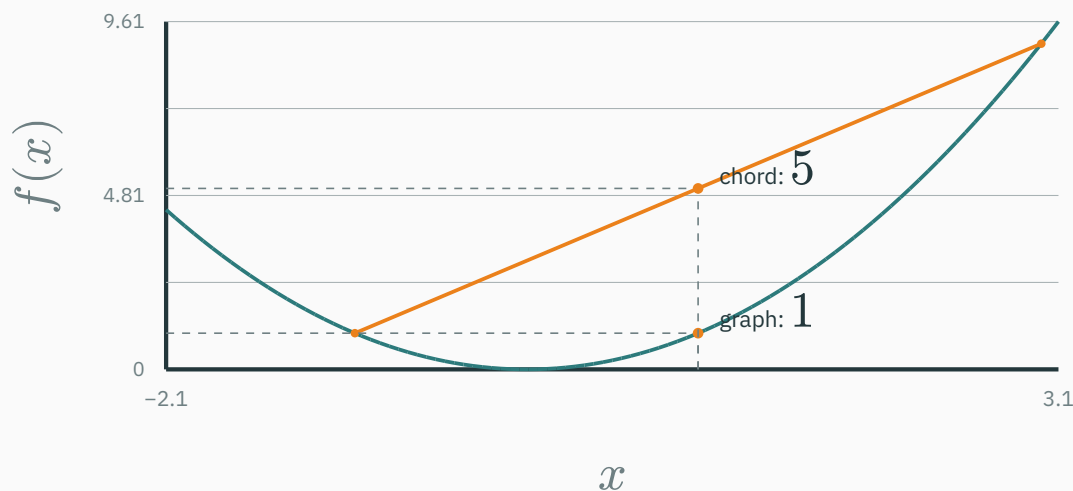
The convex sets ML lives on

Intersections preserve convexity — clip convex sets together and the result stays convex (each fence is respected). L21's LP feasible regions are exactly such stacks of half-spaces. Unions, as the two islands showed, do not.

Convex functions: the chord test

Lift the segment to the graph

A function is **convex** when every chord — the segment joining two points **on the graph** — sits on or above the graph:



$$f\left(\underbrace{tx_1 + (1-t)x_2}_{\text{mix the inputs}}\right) \leq \underbrace{tf(x_1) + (1-t)f(x_2)}_{\text{mix the outputs = chord height}} \quad \forall x_1, x_2, t \in [0, 1]$$

Convex: the graph hangs below its chords. (Domain must be a convex set
— now you know why sets came first.)

Reading the inequality with honest numbers

$f(x) = x^2$, endpoints $x_1 = -1$, $x_2 = 3$. Mixed input: $tx_1 + (1 - t)x_2 = 3 - 4t$.

t	0	1/4	1/2	3/4	1
mixed input $3 - 4t$	3	2	1	0	-1
graph $f(\text{mix}) = (3 - 4t)^2$	9	4	1	0	1
chord $tf(-1) + (1 - t)f(3) = 9 - 8t$	9	7	5	3	1

Graph $\leq$ chord in every column — with equality exactly at the endpoints. That is the definition, felt in integers.

Mnemonic: conve*x **like a cu*p** — no. Try this instead: convex holds water (U-shaped), concave is a cave (∩-shaped). Crude, universal, effective.

Two checks that need no calculus

$f(x) = x^2$, **by pure algebra** (the chord inequality, expanded):

$$\text{LHS} - \text{RHS} = (tx_1 + (1-t)x_2)^2 - tx_1^2 - (1-t)x_2^2 = -t(1-t)(x_1 - x_2)^2 \leq 0 \quad \checkmark$$

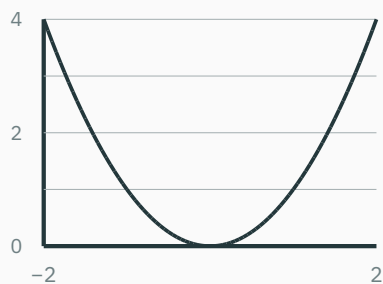
($t(1-t) \geq 0$ on $[0, 1]$; a square is ≥ 0 . Multiply out once at home — three lines.)

$f(x) = |x|$, **by the triangle inequality** (L3):

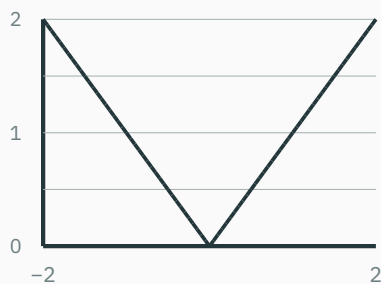
$$|tx_1 + (1-t)x_2| \leq |tx_1| + |(1-t)x_2| = t|x_1| + (1-t)|x_2| \quad \checkmark$$

The chord definition never mentions derivatives — it happily certifies the **kinked $|x|$, where $f''(0)$ does not even exist.**

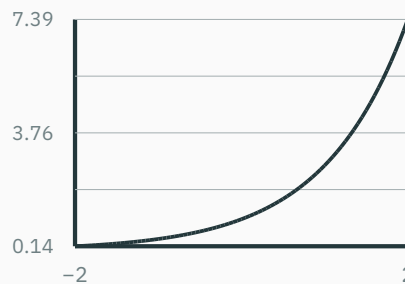
The 1-D gallery, with verdicts



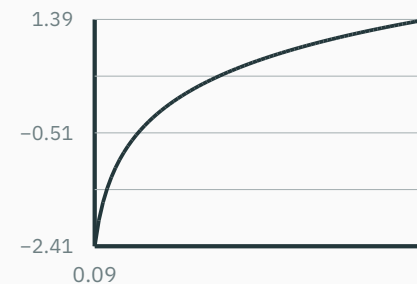
x^2 — convex on $\mathbb{R}$



$|x|$ — convex on $\mathbb{R}$

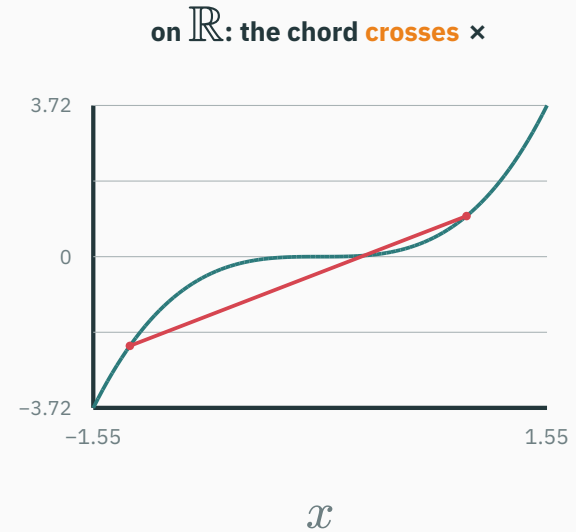
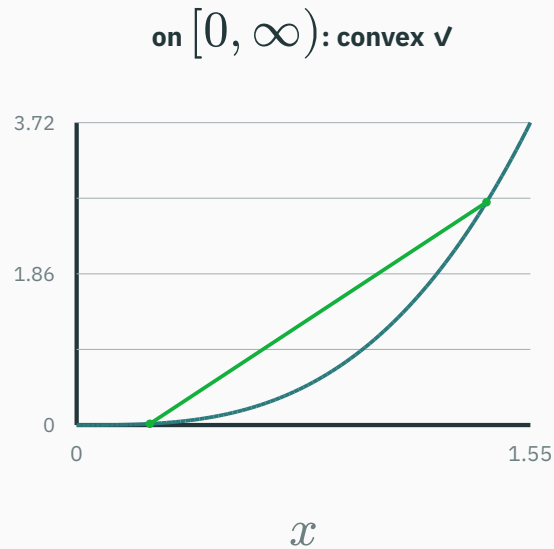


e^x — convex on $\mathbb{R}$



$\ln x$ — **concave** on $(0, \infty)$

- e^x : convex even though it never turns around — convexity is about **bending up**, not about having a minimum
- $\ln x$: bends **down** everywhere — concave; and $-\ln x$ is convex (the shape of every NLL's building block, L14)



- On $[0, \infty)$ every chord clears the graph; on $(-\infty, 0]$ it's concave; on all of $\mathbb{R}$, neither — the right chord dips below **and** rises above
- “Is f convex?” is ill-posed until you say **on which (convex) set** — same lesson the optimizers will learn about where they are allowed to search

Concave: the same story upside-down

f is **concave** $\Leftrightarrow -f$ is convex $\Leftrightarrow$ chords hang **below** the graph.

Why we care — Module 3 was secretly a concavity story:

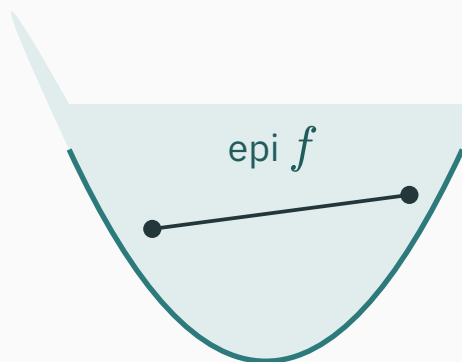
$$\underbrace{\max_{\theta} \ell(\theta)}_{\text{maximize log-likelihood (L14)}} = \underbrace{\min_{\theta} \text{NLL}(\theta)}_{\text{minimize a convex loss — often}}$$

- The coin's $\ell(\theta) = 7 \ln \theta + 3 \ln(1 - \theta)$: sum of concaves — concave; its NLL is convex
- Gaussian $\ell(\mu)$: a downward parabola in μ (L14 showed it) — concave; MSE convex

Maximizing a concave likelihood **is minimizing a convex loss. Module 3's climbs were secretly walks in a bowl.**

One idea, two costumes: the epigraph

The two halves of today are the same object. The **epigraph** of f is everything on or above the graph:



$$\text{epi } f = \{(x, v) : v \geq f(x)\}$$

- f is a convex **function** $\Leftrightarrow$ epi f is a convex **set**
- A chord's endpoints are in the epigraph, so the whole chord must be — and “chord stays in epi f ” **is** “chord above graph”

This is Boyd's favourite bridge (Ch 3.1.7) — one segment test runs the entire theory. Nothing new to memorize.

Recognizing convexity without chords

Chords are the definition — calculus is the shortcut

Checking chords for every pair is honest but slow. For twice-differentiable f , curvature settles it:

f convex on an interval $\Leftrightarrow f''(x) \geq 0$ everywhere on it

The gallery, re-certified in one line each:

f	f''	classification
x^2	$2 > 0$	convex on $\mathbb{R}$ (and strictly — no flat stretches)
e^x	$e^x > 0$	convex on $\mathbb{R}$
$\ln x$	$-1/x^2 < 0$	concave on $(0, \infty)$
x^3	$6x$ — changes sign	convex on $[0, \infty)$ only
$ x $	undefined at 0	test silent — the chord already convicted $\vee$ convex

In d dimensions: the Hessian test

f convex $\Leftrightarrow H(\theta) \succeq 0$ (PSD) for **every** θ L9's "official words", now a tool

Worked: $f(\theta_1, \theta_2) = \theta_1^2 + \theta_2^2$.

$$H = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \text{ everywhere} \Rightarrow \lambda = 2, 2 > 0 \Rightarrow \text{convex (strictly)} \checkmark$$

And the whole quadratic family at once: $q(\theta) = \frac{1}{2}\theta^\top A\theta$ has $H = A$ everywhere (L9's ★) — so **q convex $\Leftrightarrow A \succeq 0$** : bowls and troughs yes, saddles and domes no. The shape gallery was a convexity gallery all along.

The test demands PSD at **every** point of a convex domain — one PD Hessian at one point proves nothing (x^3 at $x = 1$, again).

The composition rules: convexity is stackable

Convex pieces snap together — for f, g convex:

rule	why (one breath)
αf for $\alpha \geq 0$	scaling both sides of the chord inequality preserves it
$f + g$	add the two chord inequalities
$f(Ax + b)$ — affine into convex	an affine map sends segments to segments, then f 's chords take over
$\max(f, g)$	the upper envelope: each chord clears whichever graph is on top

What the table does **not** say: products and general compositions break it. Star witness: $(x - 1)^2 \cdot (x + 1)^2 = (x^2 - 1)^2$ — two convex parabolas multiply into **our**

The composition rules: convexity is stackable

double-well mascot. Non-convexity is what multiplication (and layer-stacking!) does.

Assemble the objectives you'll actually meet

objective	build	convex?
$\ X\theta - \mathbf{y}\ ^2$ — least squares	affine $\theta \mapsto X\theta - \mathbf{y}$ into $\ \cdot\ ^2$	✓ (★ proof next section)
$\ X\theta - \mathbf{y}\ ^2 + \lambda\ \theta\ ^2$ — ridge	sum of two convex terms	✓
$\ X\theta - \mathbf{y}\ ^2 + \lambda \sum_i \theta_i $ — lasso	sum; $ \cdot $ of an affine map	✓
$\max(0, 1 - z(\theta))$, z affine — hinge	max of affine and constant	✓ (SVMs, ML course)
deep net loss	compositions upon compositions	✗ in general

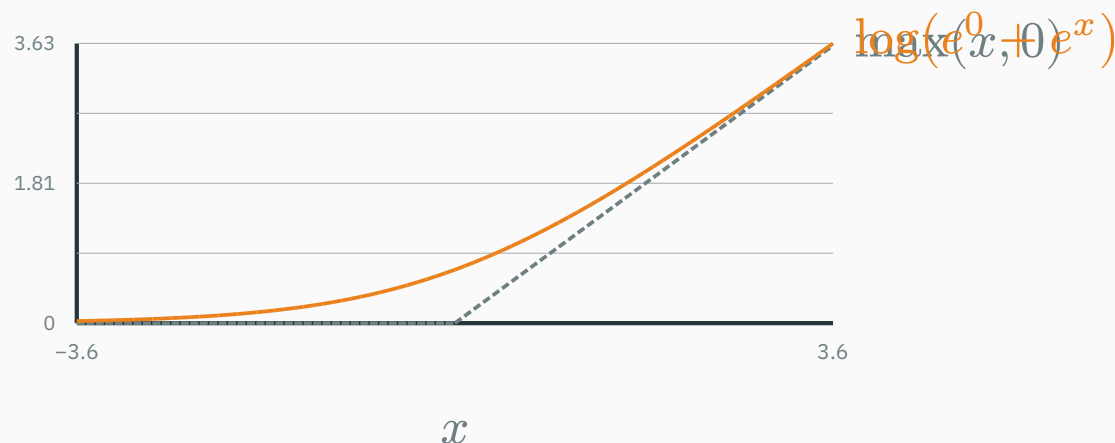
Regularizing **cannot break** convexity — L15's “every regularizer is a prior” now has an optimization corollary: priors are free, geometrically.

Assemble the objectives you'll actually meet

You will almost never chord-check a real objective. You will **assemble it from certified parts — this table is how convexity scales.**

Log-sum-exp is convex

L2 derived $\text{LSE}(x) = \log \sum_i e^{x_i}$ for numerical survival. It is also **convex** — here in 1-D against the max it approximates:



- A **smooth max**: hugs $\max(x, 0)$, never kinks — max's convexity survives the smoothing
- Why convex: its Hessian is $H = \text{diag}(p) - pp^\top$ with $p = \text{softmax}(x)$, and $v^\top H v = \text{Var}_{i \sim p}[v_i] \geq 0$ — **a variance** (L12) cannot be negative

Log-sum-exp is convex

Consequence you'll cash in ML: logistic-regression training minimizes LSE — affine, so it is convex — that's why it “just works” like least squares does.

Which one is **not** convex on all of $\mathbb{R}^2$, $\theta = (\theta_1, \theta_2)$?

A. $\|X\theta - y\|^2 + 5\|\theta\|^2$

B. $\max(\theta_1 + \theta_2, e^{\theta_1})$

C. $\log(e^{\theta_1} + e^{\theta_2})$

D. $\theta_1^2 - \theta_2^2$

Correct: D — $\theta_1^2 - \theta_2^2$ — a saddle, not a bowl

Why: Its Hessian is $\begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}$ everywhere — mixed eigenvalue signs, indefinite, the gallery's saddle: not convex (and not concave). The rest assemble from certified parts: (A) sum of convex $\checkmark$; (B) max of an affine and a convex $\checkmark$; (C) log-sum-exp $\checkmark$. Note how you convicted D and certified A–C **without one chord** — the toolkit is the point.

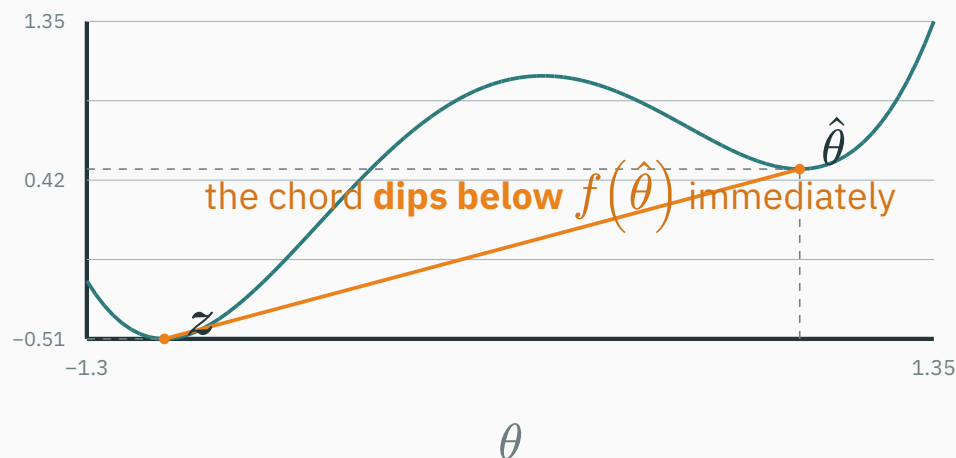
Local and global minima of convex functions

★ Every local minimum of a convex function is global

D DERIVATION

If f is convex, **every local minimum is a global minimum.**

Proof by picture first — suppose a convex f had a local min $\hat{\theta}$ **and** a strictly better point z :



★ Every local minimum of a convex function is global

D DERIVATION

The chord from $(\hat{\theta}, f(\hat{\theta}))$ to $(z, f(z))$ slopes downward. Convexity keeps the graph below that chord, so points arbitrarily close to $\hat{\theta}$ have lower values. This contradicts the assumption that $\hat{\theta}$ is a local minimum.

★ The same picture, in four lines of algebra

Setup. f convex; $\hat{\theta}$ a local min; suppose some z has $f(z) < f(\hat{\theta})$.

Walk toward z . For $t \in (0, 1]$, the mix $\theta_t = (1 - t)\hat{\theta} + tz$ lies on the segment.

Convexity:

$$f(\theta_t) \leq (1 - t)f(\hat{\theta}) + tf(z)$$

Use the bad news. Since $f(z) < f(\hat{\theta})$:

$$f(\theta_t) < (1 - t)f(\hat{\theta}) + tf(\hat{\theta}) = f(\hat{\theta})$$

Contradiction. As $t \rightarrow 0$, θ_t enters **every** ball around $\hat{\theta}$ — yet each θ_t is strictly lower. No ball certifies $\hat{\theta}$ as a local min. ■

For a convex function, a local minimum cannot coexist with a point having a lower objective value.

★ Least squares is convex

D DERIVATION

The objective L4 set up, L6 solved at scale, L14 crowned as Gaussian NLL:

$$f(\theta) = \|X\theta - \mathbf{y}\|^2 = (X\theta - \mathbf{y})^\top (X\theta - \mathbf{y}) = \theta^\top X^\top X\theta - 2\mathbf{y}^\top X\theta + \mathbf{y}^\top \mathbf{y}$$

Differentiate with L9's two earned rules ($\nabla \theta^\top A\theta = 2A\theta$ for symmetric A ; $\nabla \mathbf{b}^\top \theta = \mathbf{b}$):

$$\nabla f(\theta) = 2X^\top X\theta - 2X^\top \mathbf{y} \quad H = \nabla(\nabla f) = 2X^\top X \quad \text{--- constant: the same bowl everywhere}$$

Convexity now hangs on one question: is $X^\top X$ **positive semi-definite for every possible data matrix X** ?

★ The one-line certificate: $X^\top X \succeq 0$, always

D DERIVATION

Take any direction $v \in \mathbb{R}^d$ and test the quadratic form (L9's definition of PSD):

$$v^\top (X^\top X) v = (Xv)^\top (Xv) = \|Xv\|^2 \geq 0 \quad \checkmark$$

A squared length cannot be negative — that is the entire proof. $H = 2X^\top X \succeq 0$ for any data, so least squares is convex: its stationary point is **the global minimum.**

On our tiny dataset $x = (0, 1, 2)$, $y = (1, 2, 4)$ — everything below computed live in-slide:

$$X^\top X = \begin{pmatrix} 3 & 3 \\ 3 & 5 \end{pmatrix}, \quad \lambda = 7.16, 0.84 > 0, \quad \nabla f = 0 \Rightarrow \theta^* = \begin{pmatrix} 0.83 \\ 1.50 \end{pmatrix}$$

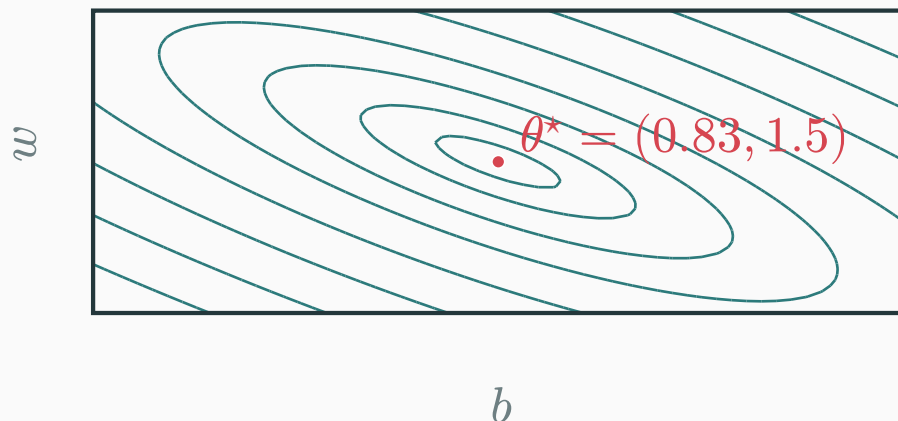
★ The one-line certificate: $X^\top X \succeq 0$, always

D DERIVATION

Both eigenvalues strictly positive $\rightarrow$ strictly convex $\rightarrow$ that θ^* ($\hat{y} = 1.5x + 0.83$) is **the unique best line in the universe** for this data. Certified.

The certified bowl, seen from above

$f(b, w) = \sum_i (b + wx_i - y_i)^2$ over the whole parameter plane — one basin, one bottom:



- Elongated, tilted — but **one** valley: every downhill path drains to θ^* (L17 walks it; the tilt and stretch will matter there)
- When columns of X are linearly dependent (L4's rank!), an eigenvalue of $X^\top X$ hits 0: the bowl grows a **flat trough** of equally-good minimizers — still convex, still all-global, just no longer unique

Compare the two fitting problems

Why does the linear model “just work”?

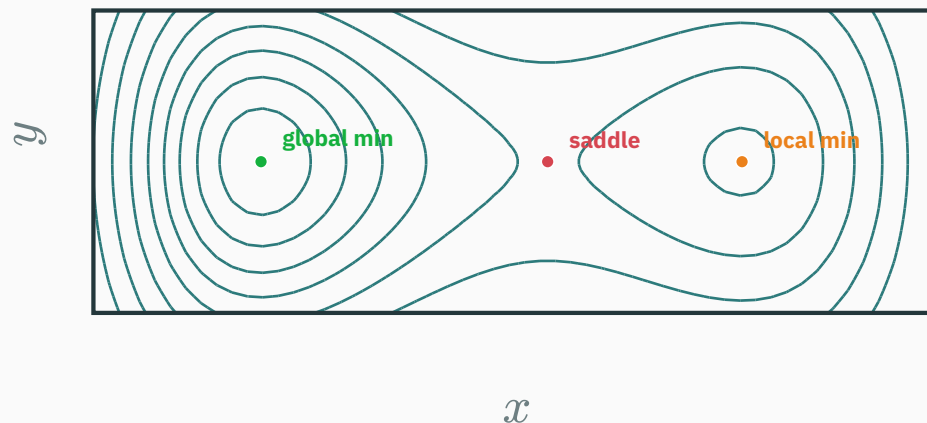
- Least squares (and ridge, lasso, logistic regression — today’s toolkit) are **convex**: one basin, local = global, any reasonable walker succeeds — no seeds, no warmup, no babysitting
- A deep network is **compositions of compositions** — the one operation the rule table refuses to bless. Its landscape genuinely holds valleys, ridges, saddles, plateaus
- So deep learning’s “tricks” are not superstition — they are **navigation gear for non-convex terrain**

Convexity is not a luxury; it is the boundary between “guaranteed” and “we’ll see”. Know which side your problem lives on.

Non-convex objectives

Several features of a non-convex objective

Our mascot's 2-D sibling, $R(x, y) = (x^2 - 1)^2 + x/2 + 1.2y^2$ — all three flavors of flat spot in one place (marks placed by the in-slide Newton solver):



- The 1-D local **max** at 0.13 became a 2-D **saddle** (down along x , up along y) — extra dimensions turn maxima into saddles
- Two basins, one ridge between: where you land is decided by where you start

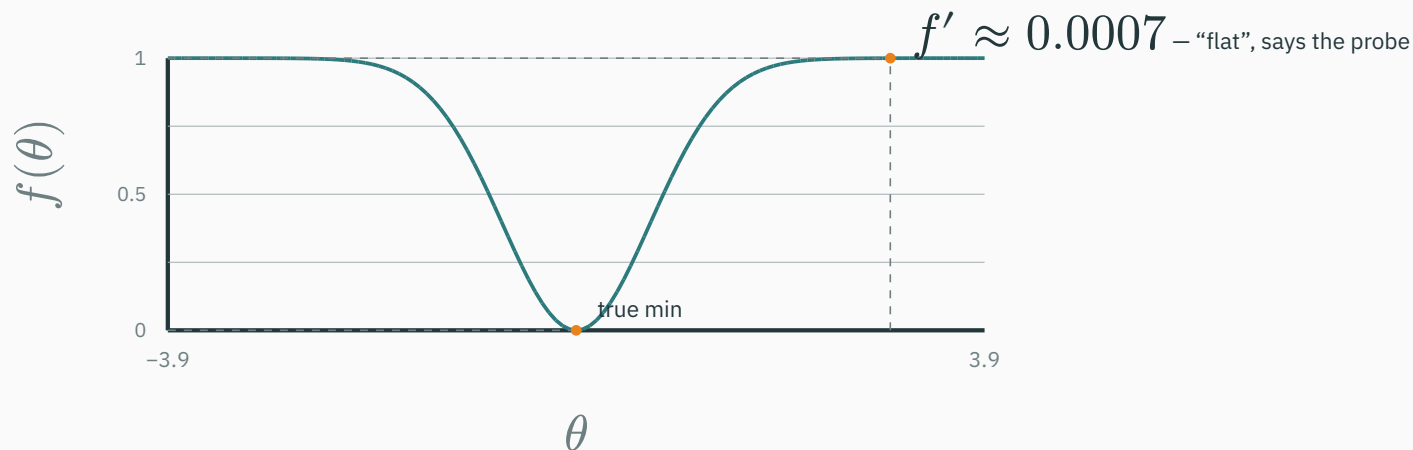
In high dimensions, saddles are the rule

Heuristic: at a random stationary point, each eigenvalue's sign is roughly a coin flip.
All d positive at once?

d	$P(\text{all } \lambda > 0) \approx 2^{-d}$	meaning
2	0.25	minima common in toy problems
20	$\approx 10^{-6}$	minima already rare
10^6 (a small net)	$\approx 10^{-301030}$	a random flat spot is a saddle, essentially always

- A saddle has a $\lambda < 0$ eigen-direction that provides a local descent direction; saddles often slow first-order methods without being local minima
- The heuristic is honest hand-waving; random-matrix theory makes it rigorous (Dauphin et al., 2014 — DL course territory)

Plateaus: small gradients away from an optimum



- On a **plateau**, $\nabla f \approx 0$ with no minimum anywhere near: steps shrink to a crawl — the failure mode behind “training is stuck” (and behind saturated activations, as the DL course will show)
- Yet deep learning works. Empirically: with enough parameters, most local minima are **nearly as good as the global one**, and modern initialization/normalization smooth the terrain. Facts used daily, proved nowhere near this course — honest IOU to ES 667

Plateaus: small gradients away from an optimum

Perspective: convexity is the **theory** of trustworthy landscapes; deep learning is the **engineering** of livable non-convex ones. You now own the vocabulary both sides negotiate in.

Lecture 16 — summary

- **One problem shape:** $\min_{\theta} f(\theta)$ — evaluated through f , ∇f , and sometimes H at the current θ .
- **Stationary points** $\nabla f = 0$: classified locally by the signs of H 's eigenvalues (L9) — minimum, maximum, or saddle.
- **Convex set:** segments never leave. **Convex function:** chords never dip below the graph; epigraph ties the two.
- **Certify fast:** $f'' \geq 0$ / $H \succeq 0$ everywhere; or assemble from rules — scale, sum, affine composition, max; log-sum-exp is convex (a variance says so).
- ★ **Convex $\Rightarrow$ local = global** — the four-line chord contradiction.
- ★ **Least squares is convex:** $H = 2X^{\top}X$ and $v^{\top}X^{\top}Xv = \|Xv\|^2 \geq 0$ — for any data.
- **Without convexity:** saddles rule high dimensions, plateaus stall walkers — deep learning's daily weather, managed by tricks, not theorems.

Lecture 16 — summary

Read before L17 — MML §7.3 to 7.3.1 (convexity, 4 pages, our notation) · Boyd & Vandenberghe, *Convex Optimization* (free PDF): **skim** §2.1–2.3 for set pictures and §3.1.1–3.1.5 + 3.2 for the function tests and operation rules — read figures and boxed statements, skip proofs on a first pass. The Distill momentum article waits for L17.

Practice problems

Try on paper; anything numeric can be checked with five lines of NumPy.

1. Chord-check $f(x) = |2x - 1|$: prove convexity from the definition (triangle inequality + the affine trick).
2. Find **all** stationary points of $f(x, y) = x^4 + y^4 - 4xy$ and classify each with the Hessian. Which are global? (Careful: there's a tie.)
3. Is $g(x) = xe^x$ convex on $\mathbb{R}$? On which interval is it convex? ($g'' = (x + 2)e^x$.)
4. Ridge: show $H = 2X^\top X + 2\lambda I$ and explain why $\lambda > 0$ makes H positive **definite** even when $X^\top X$ has a zero eigenvalue. What does that do to the trough of equally-good minimizers?
5. Prove or refute with a picture: (a) the intersection of two convex sets is convex; (b) the union is.
6. For $d = 2$, evaluate log-sum-exp's Hessian $\text{diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}^\top$ at $\mathbf{x} = (0, 0)$, find the eigenvector with $\lambda = 0$, and connect it to L2's subtract-the-max trick.

★★★ Jensen's inequality: chords, industrialized

★ OPTIONAL

Run the chord inequality with n points instead of 2 (induction), weights $t_i \geq 0$, $\sum t_i = 1$:

$$f\left(\sum_i t_i x_i\right) \leq \sum_i t_i f(x_i) \quad \text{and with } t_i = \text{probabilities: } f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)]$$

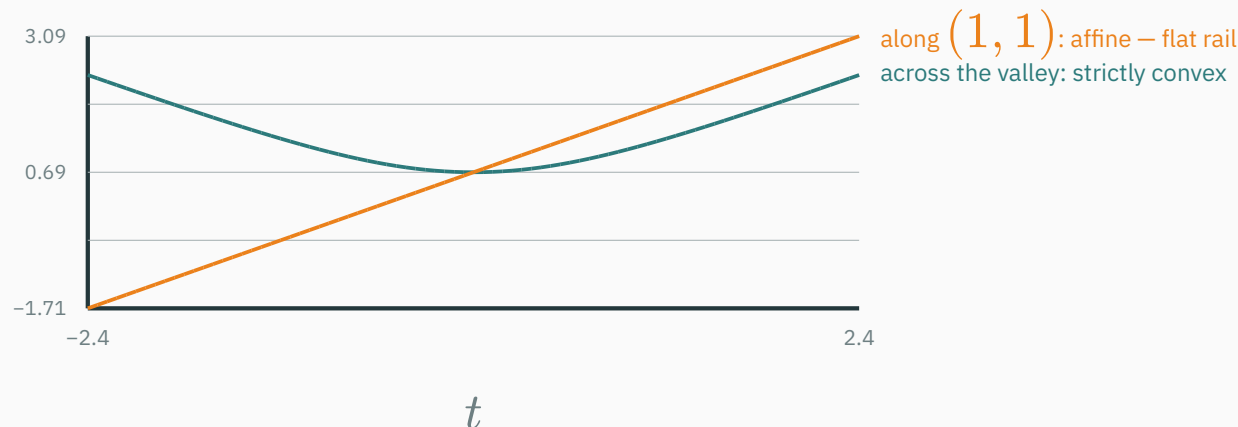
- For **concave** f , flip: $\mathbb{E}[\log X] \leq \log \mathbb{E}[X]$ — the single most-used inequality in probabilistic ML
- Module 5 runs on it: entropy bounds, $\text{KL} \geq 0$, the ELBO — all one chord, industrialized (L22–L24)

Weights on the simplex — the convex set from today's gallery. The definitions keep meeting.

☆☆☆ LSE's flat valley is L2's shift-invariance

★ OPTIONAL

Log-sum-exp is convex but not **strictly**: its bowl has one perfectly flat rail.



- Along $\mathbf{1} = (1, \dots, 1)$: $\text{LSE}(\mathbf{x} + c\mathbf{1}) = \text{LSE}(\mathbf{x}) + c$ — affine, zero curvature; indeed $H\mathbf{1} = (\text{diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}^\top)\mathbf{1} = \mathbf{p} - \mathbf{p} = \mathbf{0}$
- That flat direction **is** L2's softmax shift-invariance — subtract-the-max just slides along the rail where nothing changes. Geometry and numerics, one fact.

For a convex objective, every local minimum is global.

Next: **Gradient Descent** — now we actually walk downhill: trust the linear model for one small step, and repeat.